Journal of Scientific Computing (2023) 96:84
https://doi.org/10.1007/s10915-023-02306-2

A Physical-Constraint-Preserving Discontinuous Galerkin Method for Weakly Compressible Two-Phase Flows

Fan Zhang[1] · Jian Cheng[2] · Tiegang Liu[3]

Received: 31 December 2022 / Revised: 13 July 2023 / Accepted: 20 July 2023 /
Published online: 3 August 2023
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Abstract
This work focuses on the robust and high-order numerical simulations of weakly compressible two-phase flows by using the discontinuous Galerkin (DG) method combined with an explicit strong-stability-preserving Runge–Kutta scheme. In order to improve the computational robustness under large density ratios, a nonlinear weighted essentially non-oscillatory (WENO) limiter and a positivity-preserving limiter are specially designed and applied with the aim of dampening the nonphysical oscillations around the phase interface and preventing the occurrence of negative density, respectively. More importantly, we theoretically prove that the present method is able to satisfy the uniform-pressure–velocity criterion which states that uniform pressure and velocity profiles around an isolated phase interface should be preserved during the simulation. The performance of the present method is validated by a range of benchmark test cases with density ratios up to 1000:1. The results demonstrate that the present method possesses a good capability of simulating weakly compressible two-phase flows with large density ratios.

Keywords Weakly compressible two-phase flows · Discontinuous Galerkin method · Positivity preserving · Uniform-pressure–velocity criterion

✉ Jian Cheng
cheng_jian@iapcm.ac.cn

Fan Zhang
zhangfan0128@ustb.edu.cn

Tiegang Liu
liutg@buaa.edu.cn

1 School of Mathematics and Physics, University of Science and Technology Beijing, Beijing 100083, People's Republic of China

2 Institute of Applied Physics and Computational Mathematics, Beijing 100088, People's Republic of China

3 School of Mathematical Sciences, Beihang University, Beijing 100191, People's Republic of China

1 Introduction

Numerical simulations of two-phase flows are of great interest in many fields of science and engineering. However, since two-phase flows generally involve complex interface topology and large jumps in density [1], some specific treatments are required in developing an appropriate numerical scheme for two-phase flow simulations.

This work concerns the numerical simulation of weakly compressible two-phase flows associated with a relatively lower fluid velocity compared to the speed of sound, i.e. a relatively low Mach number. Representative applications of such kind of flows include the sloshing phenomena, breaking waves and boiling or cavitating flows, etc. First of all, it is crucial to choose an appropriate mathematical model which can describe the fluid dynamics accurately and meanwhile can be numerically solved easily. The fully compressible model [2] is a possible choice. However, since the governing equations for compressible flows become very stiff under a low Mach number, an extremely small time step is required to maintain the stability of an explicit formulation [3–7]. Another approach is to consider the truly incompressible model [8–10] which is constructed by assuming that the speed of sound is large enough to be regarded as infinite compared to the fluid velocity. The advantage of this approach is that the stringent time step restriction can be alleviated by applying a semi-implicit temporal integration scheme. However, since it needs to solve a pressure Poisson equation to enforce the incompressibility constraint, the computational costs are increased. In order to overcome the above difficulties, the weakly compressible model attracts considerable attention [11–16] due to its two favorable properties: (1) Compared to the compressible formulation, a larger time step is allowed by assuming a finite artificial speed of sound smaller than the speed of sound; (2) Compared to the incompressible formulation, the hyperbolic nature is preserved by the system of governing equations.

Based on the weakly compressible approach, Re and Abgrall [17] derived a Baer-Nunziato type model within pressure-based concept. The pressure-based formulations have also been well applied in the semi-implicit scheme for all Mach number flows [18–20]. In [21], Chanteperdrix proposed a hyperbolic three-equation model for two-phase flows which contains two mass conservation equations and one momentum conservation equation for the mixture. It is derived under the assumption that both phases are present simultaneously at any point in the solution domain with the same pressure and velocity [22]. Therefore, the phase interface is not required to be explicitly resolved but directly corresponds to the zone where the volume fraction changes rapidly from zero to one. Later, Grenier et al. [23] successfully applied the three-equation model to free-surface flow simulations. Recently, a high-order finite volume weighted essentially non-oscillatory (WENO) scheme was proposed in Ref. [24, 25] for simulating weakly compressible two-phase flows with the attempt to retain high-order accuracy and meanwhile avoid spurious oscillations.

In the past few decades, many efforts have been devoted to the development of high-order numerical schemes in the Eulerian framework for hyperbolic conservation laws [26]. Among them, the discontinuous Galerkin (DG) method, which was initially introduced by Reed and Hill [27] and further developed by Cockburn and Shu [28, 29], has been extensively applied in the area of computational fluid dynamics for its desirable properties such as local conservation, flexibility on complex geometry and high-order accuracy [30]. However, it suffers from a severe drawback that using high-order spatial approximation is prone to violate monotonicity and produce spurious oscillations which may result in nonphysical solutions and even cause a breakdown of the simulation. In order to avoid nonphysical solutions, considerable progress has been made by Zhang and Shu [31–33] in the study of

developing maximum-principle-preserving and positivity-preserving high-order numerical methods for single phase flow problems. Recently, the physical-constraint-preserving high-order numerical methods have also been successfully developed and applied to the simulations of compressible multi-medium flows [34–39].

In this work, we present a high-order DG method combined with an explicit strong-stability-preserving Runge–Kutta (SSP-RK) scheme [40] for simulating weakly compressible two-phase flows with immiscible interfaces by solving the three-equation model. More importantly, we theoretically prove that the present method possesses three compelling features. First, it satisfies the uniform-pressure–velocity (UPV) criterion [41] which states that uniform pressure and velocity profiles around an isolated phase interface should be preserved during the temporal evolution; Second, it is able to provide physical-constraint-preserving numerical solutions without visible spurious oscillations by applying a nonlinear WENO limiter [42] and a specially designed positivity-preserving limiter; Third and more importantly, it maintains the UPV criterion well after applying these two limiters. We verify the accuracy and robustness of the present method by conducting several benchmark test cases. The numerical results demonstrate that the present method is well capable of simulating weakly compressible two-phase flows with large density ratios.

The rest of this paper is arranged as follows. Section 2 describes the governing equations and the corresponding characteristic analysis. The DG discretization formulation is briefly given in Sect. 3. The nonlinear WENO limiter and the positivity-preserving limiter are introduced in detail in Sect. 4. Numerical results for both one- and two-dimensional test cases are given in Sect. 5. Finally, we provide the concluding remarks in Sect. 6.

2 Mathematical Model

2.1 Governing Equations

We consider two-dimensional weakly compressible two-phase flows with immiscible interfaces. Assuming the two phases share the same pressure and velocity, then the fluid dynamics can be described by the following model [23]

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{U}) = \mathbf{f}, \tag{1}$$

where

$$\mathbf{U} = \begin{bmatrix} \tilde{\rho}_1 \\ \tilde{\rho}_2 \\ \rho u \\ \rho v \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = [\mathbf{F}_x, \mathbf{F}_y] = \begin{bmatrix} \tilde{\rho}_1 u & \tilde{\rho}_1 v \\ \tilde{\rho}_2 u & \tilde{\rho}_2 v \\ \rho u^2 + p & \rho u v \\ \rho u v & \rho v^2 + p \end{bmatrix},$$

the flow domain is $\Omega \subset \mathbb{R}^2$, $\tilde{\rho}_k = \rho_k z_k$ is the partial density of the kth phase ($k = 1, 2$) with ρ_k and z_k being the phase density and volume fraction, respectively. Note that the admissible value of volume fraction is $z_k \in [0, 1]$ and it holds the relationship $z_1 + z_2 = 1$. $\rho = \tilde{\rho}_1 + \tilde{\rho}_2$ is the mixture density, $\mathbf{u} = [u, v]^{\mathrm{T}}$ is the velocity vector and p is the mixture pressure which is determined by the phase pressure p_k as follows

$$p = z_1 p_1 + z_2 p_2. \tag{2}$$

The righthand term in (1) is $\mathbf{f} = [0, 0, \rho \mathbf{g}]^{\mathrm{T}}$,where $\mathbf{g} = [0, -g]^{\mathrm{T}}$ with g being the gravitational acceleration.

In this work, the phase pressure p_k is obtained by the following barotropic equation of state (EoS)

$$p_k = p_0 + c_k^2 \left(\frac{\tilde{\rho}_k}{z_k} - \rho_{k,0} \right) = p_0 + c_k^2 (\rho_k - \rho_{k,0}), \tag{3}$$

where p_0 is the background pressure, c_k and $\rho_{k,0}$ are the speed of sound and reference density of the kth phase, respectively. Note that $c_k = \sqrt{\frac{\partial p_k}{\partial \rho_k}}$ is a constant and it is generally set to be at least 10 times larger than the maximum fluid velocity in order to guarantee the weakly compressible regime [23]. By imposing the instantaneous pressure equilibrium between two phases, i.e.

$$p = p_1 = p_2, \tag{4}$$

we can deduce the expression of the volume fraction as follows

$$z_1 = \frac{\gamma}{1+\gamma}, \quad z_2 = \frac{1}{1+\gamma}, \tag{5}$$

where $\gamma = \left[(q - \tilde{q}) + \sqrt{(q-\tilde{q})^2 + 4(\tilde{\rho}_1 c_1^2)(\tilde{\rho}_2 c_2^2)} \right] / (2\tilde{\rho}_2 c_2^2)$, $q = \rho_{2,0} c_2^2 - \rho_{1,0} c_1^2$ and $\tilde{q} = \tilde{\rho}_2 c_2^2 - \tilde{\rho}_1 c_1^2$. Moreover, it can be verified from (5) that given positive partial densities $\tilde{\rho}_k > 0$, it implies bounded volume fractions $z_k \in [0, 1]$.

2.2 Characteristic Analysis

The Jacobian matrix of the system of governing Eq. (1) is given by

$$\mathbf{J} = \frac{\partial (\mathbf{n}_e \cdot \mathbf{F}(\mathbf{U}))}{\partial \mathbf{U}} = \begin{bmatrix} \frac{\tilde{\rho}_2}{\rho} u_n & -\frac{\tilde{\rho}_1}{\rho} u_n & \frac{\tilde{\rho}_1}{\rho} n_x & \frac{\tilde{\rho}_1}{\rho} n_y \\ -\frac{\tilde{\rho}_2}{\rho} u_n & \frac{\tilde{\rho}_1}{\rho} u_n & \frac{\tilde{\rho}_2}{\rho} n_x & \frac{\tilde{\rho}_2}{\rho} n_y \\ \tilde{c}_1^2 n_x - u u_n & \tilde{c}_2^2 n_x - u u_n & u_n + u n_x & u n_y \\ \tilde{c}_1^2 n_y - v u_n & \tilde{c}_2^2 n_y - v u_n & v n_x & u_n + v n_y \end{bmatrix}, \tag{6}$$

where $\mathbf{n}_e = (n_x, n_y)$ is the normal vector, $u_n = \mathbf{u} \cdot \mathbf{n}_e$ is the normal velocity and

$$\tilde{c}_1^2 = \frac{\partial p}{\partial \tilde{\rho}_1} = \frac{c_1^2}{z_1} \frac{c_2^2 \tilde{\rho}_2 / z_2^2}{c_1^2 \tilde{\rho}_1 / z_1^2 + c_2^2 \tilde{\rho}_2 / z_2^2}, \quad \tilde{c}_2^2 = \frac{\partial p}{\partial \tilde{\rho}_2} = \frac{c_2^2}{z_2} \frac{c_1^2 \tilde{\rho}_1 / z_1^2}{c_1^2 \tilde{\rho}_1 / z_1^2 + c_2^2 \tilde{\rho}_2 / z_2^2}. \tag{7}$$

The Jacobian matrix (6) posses the following eigenvalues

$$\lambda_1 = u_n - c, \quad \lambda_2 = \lambda_3 = u_n, \quad \lambda_4 = u_n + c, \tag{8}$$

where c is the mixture speed of sound determined by the Wood formula, i.e.

$$\frac{1}{\rho c^2} = \frac{z_1}{\rho_1 c_1^2} + \frac{z_2}{\rho_2 c_2^2}, \tag{9}$$

which is equivalent to

$$\rho c^2 = \tilde{\rho}_1 \tilde{c}_1^2 + \tilde{\rho}_2 \tilde{c}_2^2. \tag{10}$$

From (7) to (10) we have that in order to guarantee the eigenvalues are all real values, such as to remain the system of governing Eq. (1) is hyperbolic, positive partial densities should be maintained. That is the following lemma.

Lemma 1 *Consider the system of governing Eq. (1) for weakly compressible two-phase flows, the mixture speed of sound is a real number under*

$$\tilde{\rho}_1 > 0 \quad and \quad \tilde{\rho}_2 > 0.$$

Since the system of governing Eq. (1) is hyperbolic, the high-order numerical methods for hyperbolic conservation laws can also be used for computations. In the following, we will first present a high-order discontinuous Galerkin (DG) method for solving weakly compressible two-phase flows, and then give the details on maintaining positive partial density solutions.

3 DG Formulation for Weakly Compressible Two-Phase Flows

First, we introduce some notations for convenience of exposition. Let Ω_h be an approximation of Ω, and for simplicity we divide it into a set of non-overlapping uniform rectangular cells $\Omega_h = \cup_{i=1}^{N_x} \cup_{j=1}^{N_y} I_{i,j}$. For a given cell $I_{i,j} = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [y_{j-\frac{1}{2}}, y_{j+\frac{1}{2}}]$, the cell sizes are $\Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}$, $\Delta y = y_{j+\frac{1}{2}} - y_{j-\frac{1}{2}}$, cell area is $|I_{i,j}| = \Delta x \Delta y$ and cell center is $(x_c, y_c) = ((x_{i-\frac{1}{2}} + x_{i+\frac{1}{2}})/2, (y_{j-\frac{1}{2}} + y_{j+\frac{1}{2}})/2)$. The cell edges are denoted as $\partial I_{i,j}$ and each edge is associated with a unit outward normal vector $\mathbf{n}_e = (n_x, n_y)$.

The approximate solution is defined in the following finite element space

$$V_h^K = \{v_h \in L^2(\Omega_h) : v_h|_{I_{i,j}} \in P^K(I_{i,j}), \forall I_{i,j} \in \Omega_h\}, \tag{11}$$

where $P^K(I_{i,j})$ $(K \geq 1)$ represents the space of polynomial basis functions with degree K or less on the cell $I_{i,j}$. In this work, the Legendre local orthogonal basis functions are used, e.g. the basis functions for $P^2(I_{i,j})$ are

$$\begin{aligned} &\phi_1(\mathbf{x}) = 1, \quad \phi_2(\mathbf{x}) = \frac{x - x_c}{\Delta x/2}, \ \phi_3(\mathbf{x}) = \frac{y - y_c}{\Delta y/2}, \\ &\phi_4(\mathbf{x}) = \frac{3}{2}\left(\frac{x - x_c}{\Delta x/2}\right)^2 - \frac{1}{2}, \ \phi_5(\mathbf{x}) = \frac{3}{2}\left(\frac{y - y_c}{\Delta y/2}\right)^2 - \frac{1}{2}, \ \phi_6(\mathbf{x}) = \frac{x - x_c}{\Delta x/2}\frac{y - y_c}{\Delta y/2}. \end{aligned} \tag{12}$$

Then, we can obtain a cell-wise approximation for the conservative variables

$$\mathbf{U}_h(\mathbf{x}, t) = \overline{\mathbf{U}} + \sum_{m=2}^{N_{dof}^K} \mathbf{U}^{(m)}(t)\phi_m(\mathbf{x}), \quad \forall \mathbf{x} \in I_{i,j}, \tag{13}$$

where $\mathbf{U}_h$ is the approximate solution, $\overline{\mathbf{U}} = \frac{1}{|I_{i,j}|}\int_{I_{i,j}} \mathbf{U}_h \mathrm{d}\mathbf{x}$ is the cell average, $\mathbf{U}^{(m)}(t)$ is the mth degree of freedom (DoF) at time t and N_{dof}^K is the number of DoFs. Since the solution on the interior cell edge is double-valued and discontinuous, we denote $\mathbf{U}_h^-$ and $\mathbf{U}_h^+$ which are taken from the interior and exterior of the given cell $I_{i,j}$ as the left and right limits of $\mathbf{U}_h$, respectively, and denote the traces of $\mathbf{U}_h$ on the four cell edges of $I_{i,j}$ as $\mathbf{U}_{i-\frac{1}{2},j}^-(y)$, $\mathbf{U}_{i+\frac{1}{2},j}^-(y)$, $\mathbf{U}_{i,j-\frac{1}{2}}^-(x)$ and $\mathbf{U}_{i,j+\frac{1}{2}}^-(x)$, respectively, as illustrated in Fig. 1a. Besides, along a target cell edge $\partial I_{i,j}$, we define the average and jump operators as

$$\{\{\psi_h\}\} = \frac{1}{2}(\psi_h^- + \psi_h^+), \quad [\![\psi_h]\!] = \psi_h^+ - \psi_h^-, \tag{14}$$

where ψ could be either a scalar or a vector variable.

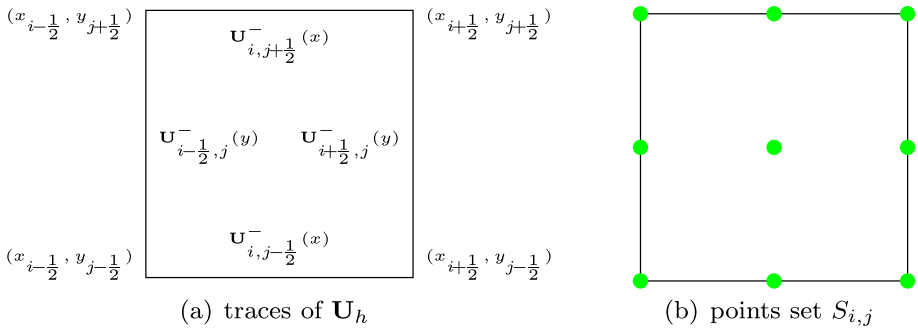


Fig. 1 The traces of solution and integral points in cell $I_{i,j}$

Multiplying the basis function ϕ_m $(1 \le m \le N^K_{dof})$ on both sides of (1), integrating over any cell $I_{i,j} \in \Omega_h$ and then integrating by parts, we obtain the DG(P^K) spatial discretization, i.e. we seek $\mathbf{U}_h \in V^K_h$ such that

$$\int_{I_{i,j}} \frac{\partial \mathbf{U}_h}{\partial t}\phi_m \mathrm{d}\mathbf{x} - \int_{I_{i,j}} \mathbf{F}(\mathbf{U}_h)\cdot\nabla\phi_m \mathrm{d}\mathbf{x} + \int_{\partial I_{i,j}} \mathbf{n}_e \cdot \widehat{\mathbf{F}}(\mathbf{U}^-_h, \mathbf{U}^+_h)\phi_m \mathrm{d}s = \int_{I_{i,j}} \mathbf{f}\phi_m \mathrm{d}\mathbf{x}, \tag{15}$$

where $\widehat{\mathbf{F}}$ is the numerical flux defined at the cell interface, e.g. the simple local Lax-Friedrichs (LLF) flux [43]

$$\mathbf{n}_e \cdot \widehat{\mathbf{F}}(\mathbf{U}^-_h, \mathbf{U}^+_h) = \mathbf{n}_e \cdot \{\{\mathbf{F}(\mathbf{U}_h)\}\} - \frac{1}{2}\alpha[\![\mathbf{U}_h]\!], \tag{16}$$

and α is the locally largest absolute eigenvalue of the Jacobian matrix (6).

In order to approximate the integrals in the DG formulation (15) accurately, the Gauss-Lobatto quadrature rule with sufficient accuracy is used for the numerical integration. We denote the one-dimensional quadrature points on the intervals $[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ and $[y_{j-\frac{1}{2}}, y_{j+\frac{1}{2}}]$ as

$$S^x_i = \{x^\alpha_i : \alpha = 1, \cdots, L\} \quad \text{and} \quad S^y_j = \{y^\beta_j : \beta = 1, \cdots, L\}, \tag{17}$$

respectively, and the quadrature weights on the interval $[-\frac{1}{2}, \frac{1}{2}]$ as w_α which satisfy $\sum_{\alpha=1}^L w_\alpha = 1$. The two-dimensional quadrature rule can be obtained from the tensor product of the one-dimensional quadrature rules as

$$S_{i,j} = S^x_i \otimes S^y_j, \tag{18}$$

see Fig. 1b for an illustration.

The temporal discretization is performed by the strong stability preserving Runge–Kutta (SSP-RK) scheme [40]. For example, the following third-order SSP-RK scheme

$$\begin{aligned}
\mathbf{U}^{n,1}_h &= \mathbf{U}^n_h + \Delta t\mathbf{R}(\mathbf{U}^n_h),\\
\mathbf{U}^{n,2}_h &= \frac{3}{4}\mathbf{U}^n_h + \frac{1}{4}\mathbf{U}^{n,1}_h + \frac{1}{4}\Delta t\mathbf{R}(\mathbf{U}^{n,1}_h),\\
\mathbf{U}^{n+1}_h &= \frac{1}{3}\mathbf{U}^n_h + \frac{2}{3}\mathbf{U}^{n,2}_h + \frac{2}{3}\Delta t\mathbf{R}(\mathbf{U}^{n,2}_h),
\end{aligned} \tag{19}$$

where the superscript 'n' denotes the time level t^n, $\Delta t = t^{n+1} - t^n$ is the time step and $\mathbf{R}(\cdot)$ is the DG spatial operator. Then, we have Theorem 1.

Theorem 1 *The DG(P^K) method* (15) *with the SSP-RK scheme* (19) *preserves uniform pressure and velocity profiles across an isolated phase interface in the absence of external forces, i.e.*

$$\textit{if}\ \ p_h^n \equiv p_0\ \ \textit{and}\ \ \mathbf{u}_h^n \equiv \mathbf{u}_0,\ \ \textit{then}\ \ p_h^{n+1} \equiv p_0\ \ \textit{and}\ \ \mathbf{u}_h^{n+1} \equiv \mathbf{u}_0,\ \ \forall I_{i,j} \in \Omega_h.$$

Proof According to (15), the DG discretization of the kth partial density is

$$L_m \frac{\mathrm{d}\tilde{\rho}_k^{(m)}}{\mathrm{d}t} - \int_{I_{i,j}} \rho_k z_k \mathbf{u}_h \cdot \nabla \phi_m \mathrm{d}\mathbf{x} + \int_{\partial I_{i,j}} \mathbf{n}_e \cdot \widehat{\rho_k z_k \mathbf{u}_h} \phi_m \mathrm{d}s = 0, \tag{20}$$

where $L_m = \int_{I_{i,j}} \phi_m \phi_m \mathrm{d}\mathbf{x}$ $(1 \le m \le N_{dof}^K)$. Since the SSP-RK scheme (19) can be regarded as a convex combination of the Euler forward scheme, we simply apply the Euler forward scheme for the temporal discretization. Besides, based on the UPV assumption and the adopted EoS (3), we have $\mathbf{u}_h^n \equiv \mathbf{u}_0$ and $\rho_k^n \equiv \rho_{k,0}$. Thus, the above equation can be further expressed as

$$(\tilde{\rho}_k^{(m)})^{n+1} = \rho_{k,0}(z_k^{(m)})^n + \frac{\Delta t}{L_m}\rho_{k,0}\Big(\int_{I_{i,j}} z_k^n \mathbf{u}_0 \cdot \nabla \phi_m \mathrm{d}\mathbf{x} - \int_{\partial I_{i,j}} \mathbf{n}_e \cdot \widehat{z_k \mathbf{u}_0} \phi_m \mathrm{d}s\Big), \tag{21}$$

where the numerical flux $\widehat{z_k \mathbf{u}_0}$ is given by

$$\mathbf{n}_e \cdot \widehat{z_k \mathbf{u}_0} = \mathbf{n}_e \cdot \mathbf{u}_0 \{\{z_k^n\}\} - \frac{1}{2}\alpha [\![z_k^n]\!]. \tag{22}$$

Then, the approximate solution of $\tilde{\rho}_k$ at the next time instant t^{n+1} is

$$\tilde{\rho}_k^{n+1} = \rho_{k,0}\left[z_k^n + \Delta t \sum_{m=1}^{N_{dof}^K} \frac{\phi_m}{L_m}\left(\int_{I_{i,j}} z_k^n \mathbf{u}_0 \cdot \nabla \phi_m \mathrm{d}\mathbf{x} - \int_{\partial I_{i,j}} \mathbf{n}_e \cdot \widehat{z_k \mathbf{u}_0} \phi_m \mathrm{d}s \right)\right]. \tag{23}$$

Substituting $\tilde{\rho}_k^{n+1}$ into the instantaneous pressure equilibrium (4), we have

$$z_k^{n+1} = z_k^n + \Delta t \sum_{m=1}^{N_{dof}^K} \frac{\phi_m}{L_m}\left(\int_{I_{i,j}} z_k^n \mathbf{u}_0 \cdot \nabla \phi_m \mathrm{d}\mathbf{x} - \int_{\partial I_{i,j}} \mathbf{n}_e \cdot \widehat{z_k \mathbf{u}_0} \phi_m \mathrm{d}s \right). \tag{24}$$

As a result, the mixture pressure at t^{n+1} is

$$p_h^{n+1} = p_0 + c_k^2 \left(\frac{\tilde{\rho}_k^{n+1}}{z_k^{n+1}} - \rho_{k,0} \right) \equiv p_0. \tag{25}$$

This implies uniform pressure can be preserved at an isolated phase interface.

Next, we consider the DG discretization of the momentum equation

$$L_m \frac{\mathrm{d}(\rho \mathbf{u})^{(m)}}{\mathrm{d}t} - \int_{I_{i,j}} (\rho_h \mathbf{u}_h \otimes \mathbf{u}_h + p_h \mathbf{I}) \cdot \nabla \phi_m \mathrm{d}\mathbf{x} + \int_{\partial I_{i,j}} \mathbf{n}_e \cdot (\widehat{\rho_h \mathbf{u}_h \otimes \mathbf{u}_h} + p_h \mathbf{I}) \phi_m \mathrm{d}s = 0. \tag{26}$$

Under uniform pressure and velocity profiles, the above equation implies

$$\begin{aligned} L_m \frac{\mathrm{d}(\rho \mathbf{u})^{(m)}}{\mathrm{d}t} &= \rho_{1,0}\mathbf{u}_0 \int_{I_{i,j}} z_1^n \mathbf{u}_0 \cdot \nabla \phi_m \mathrm{d}\mathbf{x} - \rho_{1,0}\mathbf{u}_0 \int_{\partial I_{i,j}} \mathbf{n}_e \cdot \widehat{z_1 \mathbf{u}_0} \phi_m \mathrm{d}s \\ &\quad + \rho_{2,0}\mathbf{u}_0 \int_{I_{i,j}} z_2^n \mathbf{u}_0 \cdot \nabla \phi_m \mathrm{d}\mathbf{x} - \rho_{2,0}\mathbf{u}_0 \int_{\partial I_{i,j}} \mathbf{n}_e \cdot \widehat{z_2 \mathbf{u}_0} \phi_m \mathrm{d}s \\ &= L_m \mathbf{u}_0 \left(\frac{\mathrm{d}\tilde{\rho}_1^{(m)}}{\mathrm{d}t} + \frac{\mathrm{d}\tilde{\rho}_2^{(m)}}{\mathrm{d}t}\right) = L_m \mathbf{u}_0 \frac{\mathrm{d}\rho^{(m)}}{\mathrm{d}t}. \end{aligned} \tag{27}$$

Thus, uniform velocity can also be preserved at an isolated phase interface. □

Theorem 1 illustrates that the DG method (15) for weakly compressible two-phase flows satisfies the UPV criterion which is a necessary condition for maintaining an oscillation-free phase interface.

4 Nonlinear Limiters

In this section, we will introduce two important limiters for improving the robustness of the present method in simulating weakly compressible flows with large density ratios. More importantly, we will demonstrate that the UPV criterion is still satisfied after implementing these two limiters.

4.1 WENO Limiter

In order to suppress the numerical oscillations around the phase interface, a nonlinear characteristic-wise WENO limiter [44] with a total variational bounded (TVB) type troubled-cell indicator [28] is applied during the numerical simulation. Assume the target cell is $I_{i,j}$ and the big stencils used for the WENO reconstructions in the x- and y-directions are S_x and S_y, respectively, then the implementation of the WENO limiter is given as follows.

Step 1. Calculate the right and left eigenvector matrixes in the x-direction

$$\mathbf{R}_x(\mathbf{U}) = \begin{bmatrix} \frac{\tilde{\rho}_1}{\rho} & \frac{\tilde{c}_2^2}{\tilde{c}_2^2-\tilde{c}_1^2} & 0 & \frac{\tilde{\rho}_1}{\rho} \\ \frac{\tilde{\rho}_2}{\rho} & \frac{\tilde{c}_1^2}{\tilde{c}_1^2-\tilde{c}_2^2} & 0 & \frac{\tilde{\rho}_2}{\rho} \\ u-c & u & 0 & u+c \\ v & v & 1 & v \end{bmatrix}, \quad \mathbf{L}_x(\mathbf{U}) = \begin{bmatrix} \frac{\tilde{c}_1^2+cu}{2c^2} & \frac{\tilde{c}_2^2+cu}{2c^2} & -\frac{1}{2c} & 0 \\ \frac{\tilde{\rho}_2(\tilde{c}_2^2-\tilde{c}_1^2)}{\rho c^2} & \frac{\tilde{\rho}_1(\tilde{c}_1^2-\tilde{c}_2^2)}{\rho c^2} & 0 & 0 \\ -v & -v & 0 & 1 \\ \frac{\tilde{c}_1^2-cu}{2c^2} & \frac{\tilde{c}_2^2-cu}{2c^2} & \frac{1}{2c} & 0 \end{bmatrix}. \tag{28}$$

Step 2. Project the cell-averaged conservative variables on each cell $I_{i,l} \in S_x$ to the cell-averaged characteristic variables as follows

$$\overline{\mathbf{U}}_{i,l}^{(c)} = \mathbf{L}_x(\overline{\mathbf{U}}_{i,j})\overline{\mathbf{U}}_{i,l} = [q_1, q_2, q_3, q_4]^{\mathrm{T}}. \tag{29}$$

Step 3. Apply the WENO reconstruction procedure to obtain the reconstructed polynomial solution at each quadrature point on $I_{i,j}$

$$\mathbf{U}_h^{(c)}(x_i^\alpha, y) = [\mathcal{W}_r(q_1), \mathcal{W}_r(q_2), \mathcal{W}_r(q_3), \mathcal{W}_r(q_4)]^{\mathrm{T}}. \tag{30}$$

Here, $\mathcal{W}_r(\cdot)$ denotes the WENO reconstruction procedure. For example, in the fifth-order WENO reconstruction, the big stencil $S_x = \{I_{i,l} | j-2 \le l \le j+2\}$ is generally divided

into the following three small stencils

$$S_x^{[0]} = \{I_{i,j-2}, I_{i,j-1}, I_{i,j}\},\ S_x^{[1]} = \{I_{j-1}, I_{i,j}, I_{j+1}\},\ S_x^{[2]} = \{I_{i,j}, I_{i,j+1}, I_{i,j+1}\}.$$

Then, a local quadratic polynomial $q^{[m]}(x)$ is reconstructed on $S_x^{[m]}$ $(0 \le m \le 2)$ and the WENO reconstructed polynomial solution at (x_i^α, y) is

$$\mathcal{W}_r(q) = w_\alpha^{[0]} q^{[0]}(x_i^\alpha) + w_\alpha^{[1]} q^{[1]}(x_i^\alpha) + w_\alpha^{[2]} q^{[2]}(x_i^\alpha), \tag{31}$$

where $w_\alpha^{[m]}$ is the mth nonlinear weight at (x_i^α, y) [44].

Step 4. Project the reconstructed polynomial solution back into the physical space by multiplying the cell-based right eigenvector matrix

$$\mathbf{U}_h^{(r)}(x_i^\alpha, y) = \mathbf{R}_x(\overline{\mathbf{U}}_{i,j})\mathbf{U}_h^{(c)}(x_i^\alpha, y) \tag{32}$$

Step 5. Apply the WENO reconstruction procedure in the y-direction in a similar way to obtain the reconstructed solution $\mathbf{U}_h^{(r)}(x_i^\alpha, y_j^\beta)$.

Step 6. Update the higher-order DoFs (except the cell average) of the conservative variables by using the L_2 projection

$$\widetilde{\mathbf{U}}^{(m)}(t) = \frac{1}{L_m}\sum_{\alpha=1}^{L}\sum_{\beta=1}^{L} w_\alpha w_\beta \mathbf{U}_h^{(r)}(x_i^\alpha, y_j^\beta)\phi_m(x_i^\alpha, y_j^\beta), \quad 2 \le m \le N_{dof}^K, \tag{33}$$

which finally gives the updated approximate solution as follows

$$\mathbf{U}_h^{(\text{weno})}(\mathbf{x}, t) = \overline{\mathbf{U}} + \sum_{m=2}^{N_{dof}^K} \widetilde{\mathbf{U}}^{(m)}(t)\phi_m(\mathbf{x}), \quad \forall I_{i,j} \in \Omega_h. \tag{34}$$

For the above nonlinear WENO limiter, we have the following theorem.

Theorem 2 *The nonlinear characteristic-wise WENO limiter* (34) *preserves uniform pressure and velocity profiles across an isolated phase interface, i.e.*

$$\textit{if}\ \ p_h \equiv p_0\ \ \textit{and}\ \ \mathbf{u}_h \equiv \mathbf{u}_0,\ \ \textit{then}\ \ p_h^{(weno)} \equiv p_0\ \ \textit{and}\ \ \mathbf{u}_h^{(weno)} \equiv \mathbf{u}_0,\ \ \forall I_{i,j} \in \Omega_h.$$

Proof First, under the uniform pressure assumption, the expressions of $\tilde{c}_1^2$, $\tilde{c}_2^2$ and c^2 given in (7) and (10) can be further simplified into

$$\tilde{c}_1^2 = \frac{c_1^2 c_2^2 \rho_{2,0}}{c_1^2\rho_{1,0}z_2 + c_2^2\rho_{2,0}z_1},\ \tilde{c}_2^2 = \frac{c_1^2 c_2^2 \rho_{1,0}}{c_1^2\rho_{1,0}z_2 + c_2^2\rho_{2,0}z_1},\ c^2 = \frac{c_1^2 c_2^2 \rho_{1,0}\rho_{2,0}}{\rho(c_1^2\rho_{1,0}z_2 + c_2^2\rho_{2,0}z_1)}.$$

As a consequence, the right and left eigenvector matrixes (28) are written as

$$\mathbf{R}_x(\mathbf{U}) = \begin{bmatrix} \frac{\tilde{\rho}_1}{\rho} & \frac{\rho_{1,0}}{\rho_{1,0}-\rho_{2,0}} & 0 & \frac{\tilde{\rho}_1}{\rho} \\ \frac{\tilde{\rho}_2}{\rho} & \frac{\rho_{2,0}}{\rho_{2,0}-\rho_{1,0}} & 0 & \frac{\tilde{\rho}_2}{\rho} \\ u-c & u & 0 & u+c \\ v & v & 1 & v \end{bmatrix},\ \mathbf{L}_x(\mathbf{U}) = \begin{bmatrix} \frac{\rho}{2\rho_{1,0}} + \frac{u}{2c} & \frac{\rho}{2\rho_{2,0}} + \frac{u}{2c} & -\frac{1}{2c} & 0 \\ (\frac{1}{\rho_{2,0}} - \frac{1}{\rho_{1,0}})\tilde{\rho}_2 & (\frac{1}{\rho_{1,0}} - \frac{1}{\rho_{2,0}})\tilde{\rho}_1 & 0 & 0 \\ -v & -v & 0 & 1 \\ \frac{\rho}{2\rho_{1,0}} - \frac{u}{2c} & \frac{\rho}{2\rho_{2,0}} - \frac{u}{2c} & \frac{1}{2c} & 0 \end{bmatrix},$$

and the cell averages of the characteristic variables in the x-direction are

$$\overline{\mathbf{U}}_{i,l}^{(c)} = \mathbf{L}_x(\overline{\mathbf{U}}_{i,j})\overline{\mathbf{U}}_{i,l} = \begin{bmatrix} \overline{\rho}_{i,j}/2 \\ \overline{\rho}_{i,l} - \overline{\rho}_{i,j} \\ 0 \\ \overline{\rho}_{i,j}/2 \end{bmatrix} = \begin{bmatrix} [q_1]_{i,j} \\ [q_2]_{i,l} \\ 0 \\ [q_1]_{i,j} \end{bmatrix}, \tag{35}$$

where $[\cdot]_{i,l}$ denotes the corresponding cell-averaged value on $I_{i,l} \in S_x$.

Next, for each quadrature point $x_i^\alpha \in S_i^x$, by applying the WENO reconstruction procedure and then projecting the reconstructed solution back into the physical space, it gives

$$\begin{aligned}\mathbf{U}_h^{(r)}(x_i^\alpha, y) &= \mathbf{R}_x(\overline{\mathbf{U}}_{i,j})\mathbf{U}_h^{(c)}(x_i^\alpha, y) \\ &= \begin{bmatrix} 2[\tilde{\rho}_1/\rho]_{i,j}\mathcal{W}_r([q_1]_{i,j}) + \frac{\rho_{1,0}}{\rho_{1,0}-\rho_{2,0}}\mathcal{W}_r([q_2]_{i,l}) \\ 2[\tilde{\rho}_2/\rho]_{i,j}\mathcal{W}_r([q_1]_{i,j}) - \frac{\rho_{2,0}}{\rho_{1,0}-\rho_{2,0}}\mathcal{W}_r([q_2]_{i,l}) \\ 2u_0\mathcal{W}_r([q_1]_{i,j}) + u_0\mathcal{W}_r([q_2]_{i,l}) \\ 2v_0\mathcal{W}_r([q_1]_{i,j}) + v_0\mathcal{W}_r([q_2]_{i,l}) \end{bmatrix}. \end{aligned} \tag{36}$$

It can be verified that the reconstructed pressure and velocity profiles are equal to p_0 and $\mathbf{u}_0$, respectively, see Appendix A for a detailed proof. In addition, the same result can also be achieved by $\mathbf{U}_h^{(r)}(x_i^\alpha, y_j^\beta)$.

Finally, we update the high-order DoFs based on the WENO reconstruction polynomial solution $\mathbf{U}_h^{(r)}(x_i^\alpha, y_j^\beta)$. Then, the corresponding updated approximate solution of the partial density on the target cell $I_{i,j}$ is

$$\tilde{\rho}_k^{(\text{weno})} = \overline{(\tilde{\rho}_k)} + \sum_{m=2}^{N_{dof}^K} \frac{\phi_m}{L_m}\left[\sum_{\alpha=1}^{L}\sum_{\beta=1}^{L} w_\alpha w_\beta \tilde{\rho}_k^{(r)}(x_i^\alpha, y_j^\beta)\phi_m(x_i^\alpha, y_j^\beta)\right], \tag{37}$$

which gives the updated equilibrium volume fraction as

$$z_k^{(\text{weno})} = \tilde{\rho}_k^{(\text{weno})}/\rho_{k,0}.$$

Therefore, the updated mixture pressure is

$$p_h^{(\text{weno})} = p_0 + c_k^2\left(\frac{\tilde{\rho}_k^{(\text{weno})}}{z_k^{(\text{weno})}} - \rho_{k,0}\right) \equiv p_0.$$

Similarly, the updated solution of velocity is calculated by

$$\begin{aligned}\mathbf{u}_h^{(\text{weno})} &= \frac{(\rho\mathbf{u})_h^{(\text{weno})}}{\tilde{\rho}_1^{(\text{weno})} + \tilde{\rho}_2^{(\text{weno})}} \\ &= \frac{\overline{(\rho\mathbf{u})} + \sum_{m=2}^{N_{dof}^K} \frac{\phi_m}{L_m}\Big[\sum_{\alpha=1}^{L}\sum_{\beta=1}^{L} w_\alpha w_\beta (\rho\mathbf{u})_h^{(r)}(x_i^\alpha, y_j^\beta)\phi_m(x_i^\alpha, y_j^\beta)\Big]}{\overline{\rho} + \sum_{m=2}^{N_{dof}^K} \frac{\phi_m}{L_m}\Big[\sum_{\alpha=1}^{L}\sum_{\beta=1}^{L} w_\alpha w_\beta \rho_h^{(r)}(x_i^\alpha, y_j^\beta)\phi_m(x_i^\alpha, y_j^\beta)\Big]} \\ &\equiv \mathbf{u}_0.\end{aligned}$$

This completes the proof. □

Theorem 2 guarantees that the UPV criterion can still be ensured after implementing the nonlinear WENO limiter to the DG approximate solution.

4.2 Positivity-Preserving Limiter

As mentioned in Lemma 1, in order to maintain the hyperbolic nature of the system of governing equations, the partial densities should remain positive. For this purpose, we design a positivity-preserving limiter and apply it to the DG approximate solution in this section.

According to (15), the DG discretization of the cell-averaged conservative variable vector is given by

$$\begin{aligned}\overline{\mathbf{U}}^{n+1} = \overline{\mathbf{U}}^{n} &- \frac{\lambda_x}{\Delta y}\int_{y_{j-\frac{1}{2}}}^{y_{j+\frac{1}{2}}} \widehat{f}\big(\mathbf{U}^-_{i+\frac{1}{2},j}(y), \mathbf{U}^+_{i+\frac{1}{2},j}(y)\big) - \widehat{f}\big(\mathbf{U}^+_{i-\frac{1}{2},j}(y), \mathbf{U}^-_{i-\frac{1}{2},j}(y)\big)\mathrm{d}y \\ &- \frac{\lambda_y}{\Delta x}\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \widehat{g}\big(\mathbf{U}^-_{i,j+\frac{1}{2}}(x), \mathbf{U}^+_{i,j+\frac{1}{2}}(x)\big) - \widehat{g}\big(\mathbf{U}^+_{i,j-\frac{1}{2}}(x), \mathbf{U}^-_{i,j-\frac{1}{2}}(x)\big)\mathrm{d}x + \Delta t\overline{\mathbf{f}}^n,\end{aligned} \tag{38}$$

where $\lambda_x = \Delta t/\Delta x$, $\lambda_y = \Delta t/\Delta y$, $\widehat{f}(\cdot,\cdot)$ and $\widehat{g}(\cdot,\cdot)$ are the corresponding one-dimensional LLF fluxes in the x- and y-directions, respectively. Then, we have the following theorem.

Theorem 3 *Consider the scheme satisfied by the cell averages of DG(P^K) method* (38) *for weakly compressible two-phase flows. If the approximate polynomial solution satisfies* $\tilde{\rho}_k(\mathbf{x}, t^n) > 0$ *for any point in the quadrature set* $S_{i,j}$*, then* $\overline{(\tilde{\rho}_k)}^{n+1} > 0$ *under the time step restriction*

$$(\lambda_x + \lambda_y)\||\mathbf{u}| + c\|_\infty \le \min\{w_1, 1/(2K+1)\}, \tag{39}$$

where w_1 *is the Gauss-Lobatto quadrature weight and* $1/(2K+1)$ *is introduced for maintaining the linear stability requirement for the DG(P^K) method* [28].

Proof The proof is similar to that of Theorem 2.6 in [33], and thus omitted. □

With positive cell-averaged partial density $\overline{(\tilde{\rho}_k)}$, we also need to modify the whole polynomial solution $\tilde{\rho}_k$ such that $\tilde{\rho}_k > 0$ for all quadrature points within the set $S_{i,j}$. This requirement can be enforced by employing a linear scaling limiter to $\tilde{\rho}_k$ as follows [45]

$$\tilde{\rho}_k^* = \overline{(\tilde{\rho}_k)} + \theta[\tilde{\rho}_k - \overline{(\tilde{\rho}_k)}], \quad \forall I_{i,j} \in \Omega_h, \tag{40}$$

where θ is the scaling factor defined as

$$\theta = \min_{k=1,2}\theta_k, \quad \theta_k = \min\left\{\frac{\overline{(\tilde{\rho}_k)} - \varepsilon}{\overline{(\tilde{\rho}_k)} - m_k}, 1\right\}, \quad m_k = \min_{(x,y)\in S_{i,j}}\tilde{\rho}_k, \tag{41}$$

and $\varepsilon = \min\{10^{-13}, \overline{(\tilde{\rho}_k)}\}$ is a small positive constant.

Similarly, after implementing the positivity-preserving limiter to the DG approximate solution, we need to demonstrate that the UPV criterion can still be preserved. By substituting the modified partial density (40) into the pressure equilibrium equation

$$p_0 + c_1^2\left(\frac{\tilde{\rho}_1^*}{z_1^*} - \rho_{1,0}\right) = p_0 + c_1^2\left(\frac{\tilde{\rho}_2^*}{z_2^*} - \rho_{2,0}\right), \tag{42}$$

we have the modified volume fraction z_k^*

$$z_k^* = \tilde{\rho}_k^*/\rho_{k,0}, \tag{43}$$

which satisfies

$$z_1^* + z_2^* = 1 \quad \text{and} \quad z_k^* > 0, \quad k = 1, 2. \tag{44}$$

Moreover, since

$$p_0 + c_k^2\left(\frac{\tilde{\rho}_k^*}{z_k^*} - \rho_{k,0}\right) \equiv p_0, \tag{45}$$

uniform pressure is preserved. Next, in order to preserve uniform velocity, we correct the polynomial solution of the conservative variable $\rho\mathbf{u}$ as

$$(\rho\mathbf{u})_h^* = \left(\tilde{\rho}_1^* + \tilde{\rho}_2^*\right)\mathbf{u}_h, \tag{46}$$

and update higher order DoFs (except the cell average) by the L_2 projection

$$(\rho\mathbf{u})^{(m)}(t) = \frac{1}{L_m}\int_{I_{i,j}} (\rho\mathbf{u})_h^*\phi_m \mathrm{d}\mathbf{x}, \quad 2 \le m \le N_{dof}^K. \tag{47}$$

As a consequence, we have the following theorem

Theorem 4 *The positivity-preserving limiter* (40) *preserves uniform pressure and velocity profiles across an isolated phase interface, i.e.*

$$\textit{if} \;\; p_h \equiv p_0 \;\; \textit{and} \;\; \mathbf{u}_h \equiv \mathbf{u}_0, \;\; \textit{then} \;\; p_h^* \equiv p_0 \;\; \textit{and} \;\; \mathbf{u}_h^* \equiv \mathbf{u}_0, \;\; \forall I_{i,j} \in \Omega_h,$$

where the modified approximate solutions are given by (40) *and* (46).

4.3 Summary

Finally, we summary the overall limiting algorithm of the present method in Algorithm 1. For a SSP-RK temporal discretization scheme, the limiting procedure should be employed for each stage.

Algorithm 1 The nonlinear limiting algorithm.

Input: The original DG solution $\mathbf{U}_h^{(n0)}(\mathbf{x}, t)$
Output: The modified DG solution $\mathbf{U}_h(\mathbf{x}, t)$

```
if cell I_{i,j} is a troubled cell then
    apply WENO limiter in Sect. 4.1 to obtain the solution U_h^{(n1)}(x, t)
else
    U_h^{(n1)}(x, t) = U_h^{(n0)}(x, t)
end if
if ρ̃_k^{(n1)}(x_i^α, y_j^β) < ε, (x_i^α, y_j^β) ∈ S_{i,j} then
    apply positivity-preserving limiter in Sect. 4.2 to obtain the solution U_h^{(n2)}(x, t)
else
    U_h^{(n2)}(x, t) = U_h^{(n1)}(x, t)
end if
U_h(x, t) = U_h^{(n2)}(x, t)
```

5 Numerical Tests

In this section, five benchmark problems are tested to verify the accuracy and robustness of the present method. For simplicity, we denote the DG method with the WENO limiter, the positivity-preserving limiter and them both as DG-WENO, DG-PP and DG-WENO-PP methods, respectively. Unless otherwise stated, the time step is taken based on the time step restriction (39).

Table 1 Convergence order for the 1D smooth interface advection problem

Cells	DG(P^1)				DG(P^2)			
	L_1-norm	L_1-order	L_2-norm	L_2-order	L_1-norm	L_1-order	L_2-norm	L_2-order
40	4.42e–1		5.18e–1		1.96e–2		2.84e–2	
80	1.08e–1	2.03	1.27e–1	2.03	2.73e–3	2.84	3.94e–3	2.85
160	2.69e–2	2.00	3.16e–2	2.01	3.58e–4	2.93	5.12e–4	2.94
320	6.70e–3	2.00	7.90e–3	2.00	4.59e–5	2.96	6.53e–5	2.97

5.1 One-Dimensional Interface Advection Problem

The first case is a one-dimensional smooth interface advection test initialized with a uniform velocity field $u(x, 0) = 1\text{m/s}$ and a smooth volume fraction

$$z_1(x, 0) = (0.5 - \varepsilon_z)\sin(2\pi x) + 0.5, \quad x \in [0, L], \tag{48}$$

where $\varepsilon_z = 10^{-2}$ and $L = 1$ m is the length of the computational domain. The initial conditions for the partial densities are set to be

$$\rho_1(x, 0) = \rho_{1,0} = 1000\,\text{kg/m}^3, \quad \rho_2(x, 0) = \rho_{2,0} = 1\,\text{kg/m}^3, \tag{49}$$

which leads to a uniform pressure distribution $p(x, 0) = p_0 = 0$ Pa. The artificial speeds of sound are taken as $c_1 = c_2 = 70$ m/s. Periodic boundary conditions are imposed and the CFL number is 0.18. The computations are conducted at four different mesh resolutions with the number of cells $N = 40, 80, 160, 320$ in order to study the convergence property of the present method. Since the solution is smooth enough, no limiting procedure is applied.

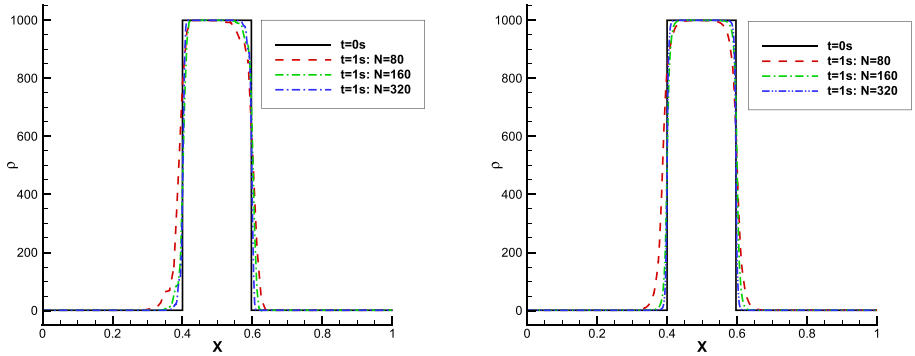
The convergence order for the one-dimensional smooth interface advection problem obtained by the DG(P^1) and DG(P^2) methods at $t = 2$ s are listed in Table 1. Here, the L_1-norm and L_2-norm are used to evaluate the discrete error magnitudes of the mixture density ρ. It can be seen that the present DG(P^K) method can achieve the optimal $(K + 1)$-th order of convergence for solving the one-dimensional smooth interface advection problem.

Next, we consider a sharp initial volume fraction given as follows

$$z_1(x, 0) = \begin{cases} 1 - \varepsilon_z, & x \in \left(\dfrac{L}{2} - r_0, \dfrac{L}{2} + r_0\right), \\ \varepsilon_z, & \text{otherwise}, \end{cases} \tag{50}$$

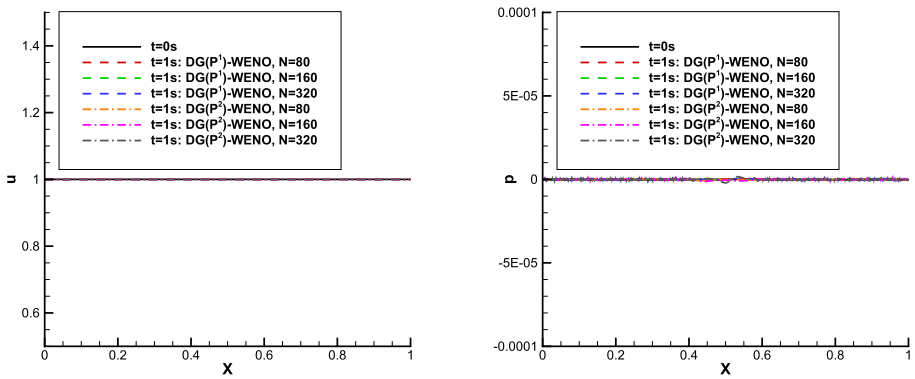
where $\varepsilon_z = 10^{-4}$ and $r_0 = L/10$ is the half length of the interface. The output time is $t = 1$ s and all the other variables remain the same as in the previous smooth interface advection problem. This case is used to verify the ability of the present method in maintaining the oscillation-free property of velocity and pressure at an isolated phase interface. Since the solution contains sharp gradient regions, the nonlinear WENO limiter is applied in this test case.

In Fig. 2 we compare the numerical solutions of the mixture density obtained by the DG-WENO method with different polynomial approximation orders and mesh resolutions. We can observe that the initial sharp interface becomes smeared as time evolves. While with the increase of the polynomial approximation order and the number of mesh cells, the numerical solutions are getting closer to the initial solution. In order to verify the oscillation-free property of the proposed method, we present the numerical solutions and the oscillation



(a) DG(P^1)-WENO method (b) DG(P^2)-WENO method

Fig. 2 Density for the 1D sharp interface advection problem ($\varepsilon_z = 10^{-4}$)

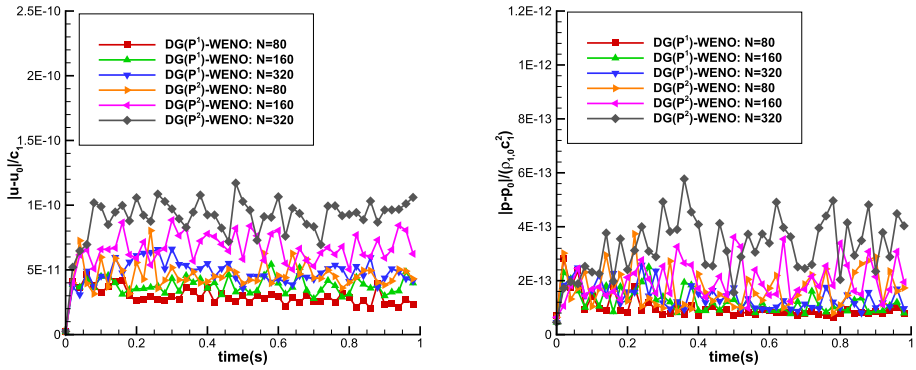


(a) velocity field (b) pressure field

Fig. 3 Velocity and pressure for the 1D sharp interface advection problem ($\varepsilon_z = 10^{-4}$)

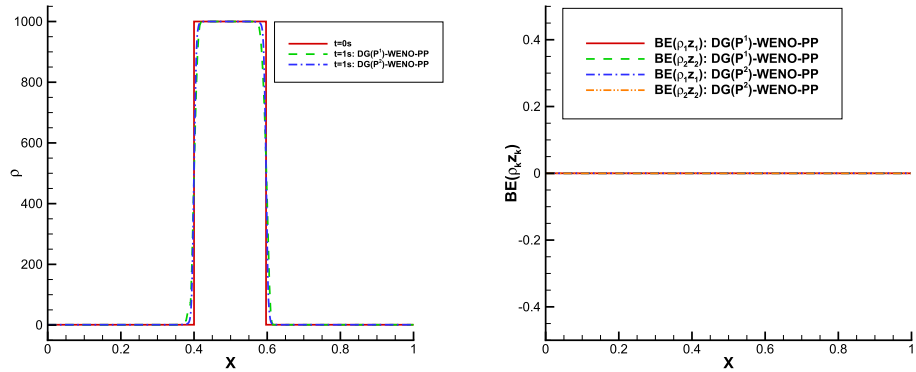
evolutions of velocity and pressure in Figs. 3 and 4, respectively. It is obvious to see that uniform velocity and pressure fields with nearly zero oscillation, e.g. around $O(10^{-10})$ and $O(10^{-13})$ for velocity and pressure, respectively, can be ensured by the present DG-WENO method.

Finally, we test the feasibility of the positivity-preserving limiter. In order to initialize the lower and upper bounds of the volume fraction more close to 0 and 1, respectively, we decrease ε_z to 10^{-8} in (50). Under this case, since the high-order DG method violates the monotonicity, negative partial density appears in the vicinity of the phase interface which leads to a breakdown during the simulation. Thus, the DG-WENO-PP method is applied in this test case for improving robustness. Figure 5a shows the mixture density obtained by the DG-WENO-PP method with 320 uniform cells. We can observe that the positivity-preserving limiter improves the computational robustness significantly and the present DG-WENO-PP method can capture the high and low mixture density regions accurately. In order to estimate the potential undershoots near the phase interface, we introduce the bound error for the partial



(a) velocity oscillation (b) pressure oscillation

Fig. 4 Oscillation evolution for the 1D sharp interface advection problem ($\varepsilon_z = 10^{-4}$)



(a) mixture density (b) bound error

Fig. 5 Density and bound error for the 1D sharp interface advection problem ($\varepsilon_z = 10^{-8}$)

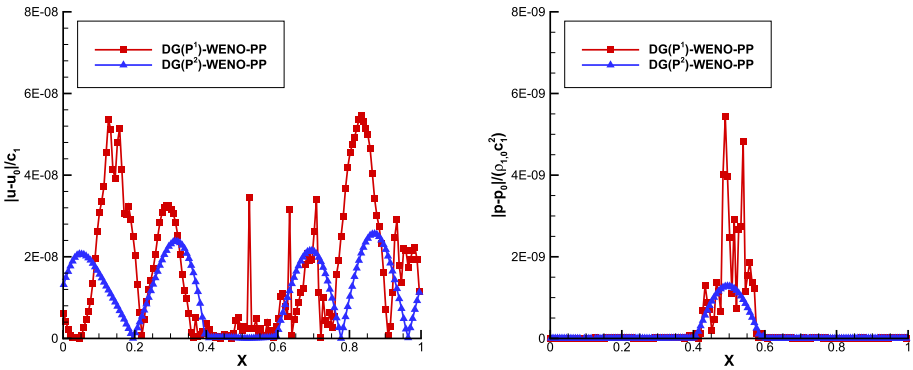
density as follows

$$\mathrm{BE}(\rho_k z_k) = \begin{cases} \left|\overline{(\rho_k z_k)}\right|, & \text{if} \quad \rho_k z_k < 0, \\ 0, & \text{otherwise.} \end{cases} \tag{51}$$

The numerical result for $\mathrm{BE}(\rho_k z_k)$ at $t = 1$ s is shown in Fig. 5b which verifies that the positivity of the partial density is well preserved by the positivity-preserving limiter. The velocity and pressure oscillation distributions at $t = 1$ s are shown in Fig. 6. As expected, the present method can still keep the oscillation-free property well after applying the positivity-preserving limiter.

5.2 Two-Phase Vortex Advection Problem

In this case, we test the performance of the present method in solving two-dimensional two-phase vortex advection problem. The initial conditions within the periodic computational



(a) velocity oscillation (b) pressure oscillation

Fig. 6 Oscillation distribution for the 1D sharp interface advection problem ($\varepsilon_z = 10^{-8}$)

domain $[-L, L]^2$ are imposed as [24]

$$\begin{cases} z_1(r,\theta,0) = (1-2\varepsilon_z)\exp(-\dfrac{r^2}{r_0^2}) + \varepsilon_z, \\ p(r,\theta,0) = P\left(1-\exp(-\dfrac{r^2}{r_0^2})\right), \\ u(r,\theta,0) = U - \sin\theta\sqrt{\dfrac{r}{\rho}\dfrac{\partial p}{\partial r}}, \\ v(r,\theta,0) = V + \cos\theta\sqrt{\dfrac{r}{\rho}\dfrac{\partial p}{\partial r}}, \end{cases} \tag{52}$$

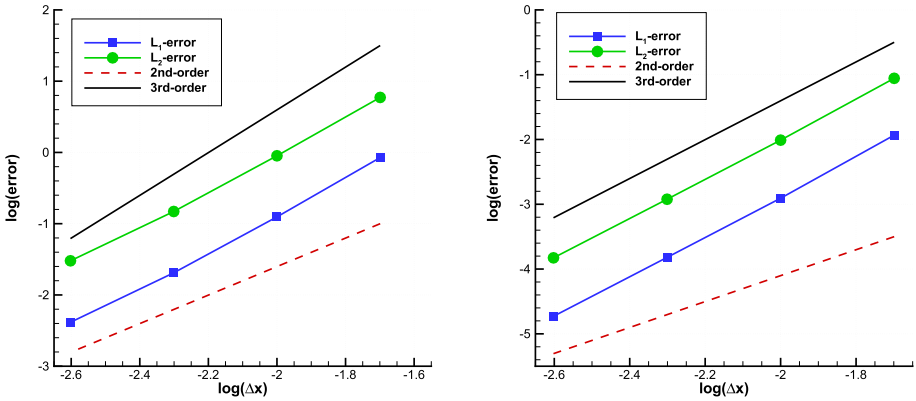
where r and θ are the cylindrical coordinates with $x = r\cos\theta$ and $y = r\sin\theta$, $L = 1$ m, $r_0 = 0.2L$ and $\varepsilon_z = 10^{-1}$. The density ratio is $\rho_{1,0}/\rho_{2,0} = 1000$, the speed of sound is $c_1 = c_2 = 30$ m/s, $P = 0.1\rho_{2,0}c_2^2$ and $U = V = 2$ m/s are all constants. The final time is $t = 1$ s and the CFL number is 0.18. The computations are conducted on four different mesh resolutions with the number of cells taken as $N = 50^2$, 100^2, 200^2 and 400^2.

In Fig. 7, we plot the computational accuracy which indicates that the present DG(P^K) method is able to achieve the optimal $(K+1)$-th convergence order for the two-phase vortex advection problem. The initial and final mixture density fields obtained by the DG(P^2) method on the finest mesh is shown in Fig. 8. It can be seen that the final result exhibits a good agreement with the initial solution.

5.3 Low Amplitude Standing Wave

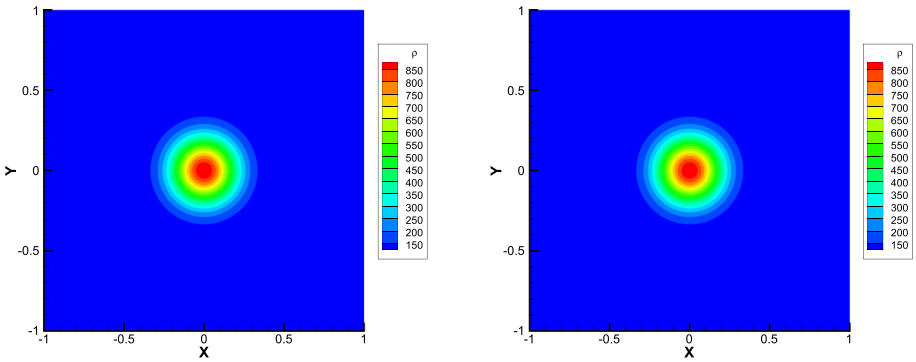
In this test, we consider the low amplitude standing wave problem [10] to validate the ability of the present method in simulating free-surface hydrodynamic problems. At $t = 0$s, both the air and water are at rest under the influence of gravity in a square container $\Omega = [0, 0.5\text{m}]^2$ which is composed of four free-slip solid walls. The air is superposed to the water and the initial interface is

$$y(\mathbf{x}, 0) = h_s + A\cos(k_n x), \tag{53}$$



(a) DG(P^1) method (b) DG(P^2) method

Fig. 7 Convergence order for the two-phase vortex advection problem

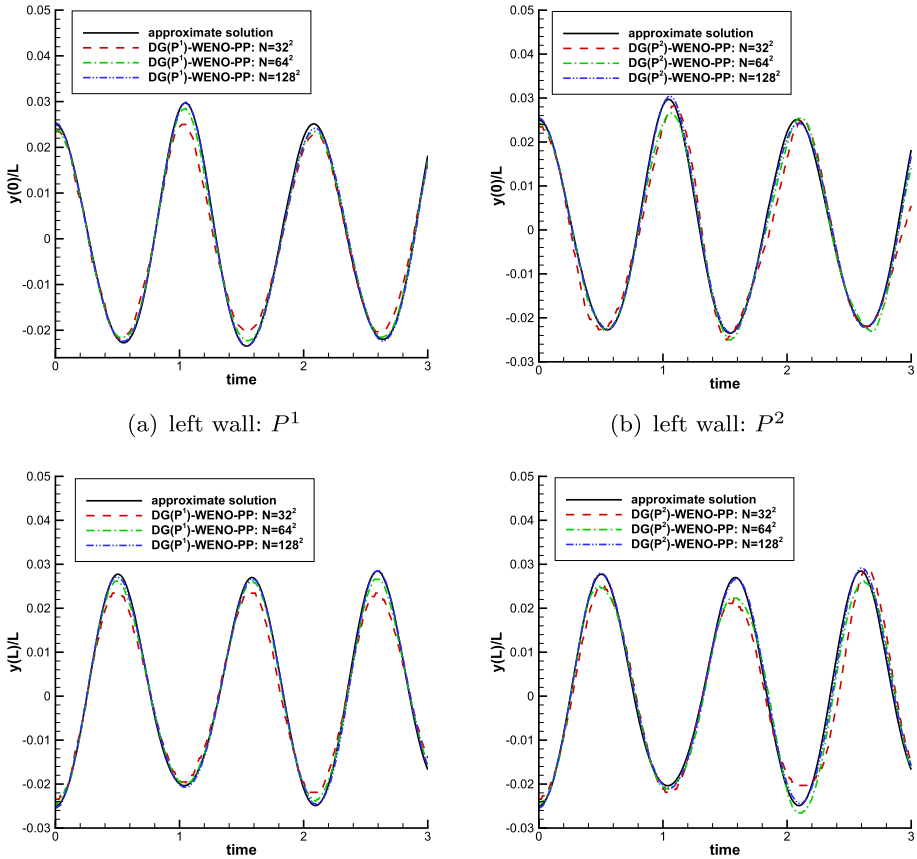


(a) $t = 0$s (b) $t = 1$s

Fig. 8 Initial and final density fields for the two-phase vortex advection problem

where $h_s = 0.25$m is the still water depth, $A = 0.025$m is the initial wave amplitude and $k_n = 2\pi$ is the wave number. The gravity is $\mathbf{g} = (0, -2\pi \text{ m/s}^2)^{\mathrm{T}}$, the density ratio is 1000 with $\rho_{1,0} = 1000$ kg/m^3 and $\rho_{2,0} = 1$ kg/m^3. The speed of sound for both phases are set as $c_1 = c_2 = 100$ m/s and the background pressure is $p_0 = 10{,}000$ Pa. As described in Ref. [46], the predictions of the low amplitude free-surface motions can be obtained by applying the second-order small perturbation theory. An approximate expression of the oscillation period is $T = 2\pi/\sqrt{|g|k_n \tanh(k_n h_s)} \simeq 1.04$ s [47].

In Fig. 9, we compare the predicted and the computed interface evolutions for this test, where the computations are conducted by the DG-WENO-PP method with three different mesh resolutions $N = 32^2, 64^2$ and 128^2. As we can see, the computed results exhibit a good agreement with the prediction solution as the number of mesh cells is increased. Table 2 lists the computed time period of oscillation which is also very close to the reference one.



(a) left wall: P^1 (b) left wall: P^2

(c) right wall: P^1 (d) right wall: P^2

Fig. 9 Interface evolution for the low amplitude standing wave problem

Table 2 Time period for the low amplitude standing wave problem (s)

Cells	32×32	64×64	128×128
DG(P^1)-WENO-PP	1.00	1.03	1.05
DG(P^2)-WENO-PP	1.07	1.06	1.05

5.4 Rayleigh–Taylor Instability

In order to demonstrate the performance of the present method in simulating flows involving complex interfaces, the Rayleigh–Taylor instability problem [48] is considered in this test case. It consists of two immiscible flows of different densities, with the heavier fluid of density $\rho_{1,0} = 1.225\text{kg/m}^3$ kept at the top and the lighter fluid of density $\rho_{2,0} = 0.1694\text{kg/m}^3$ kept at the bottom of a rectangular container $\Omega = [0, 1\text{m}] \times [0, 4\text{m}]$. Initially, the velocity field is set to be zero and the position of the perturbed interface between two fluids is

$$y(\mathbf{x}, 0) = 2 + 0.05\cos(2\pi x). \tag{54}$$

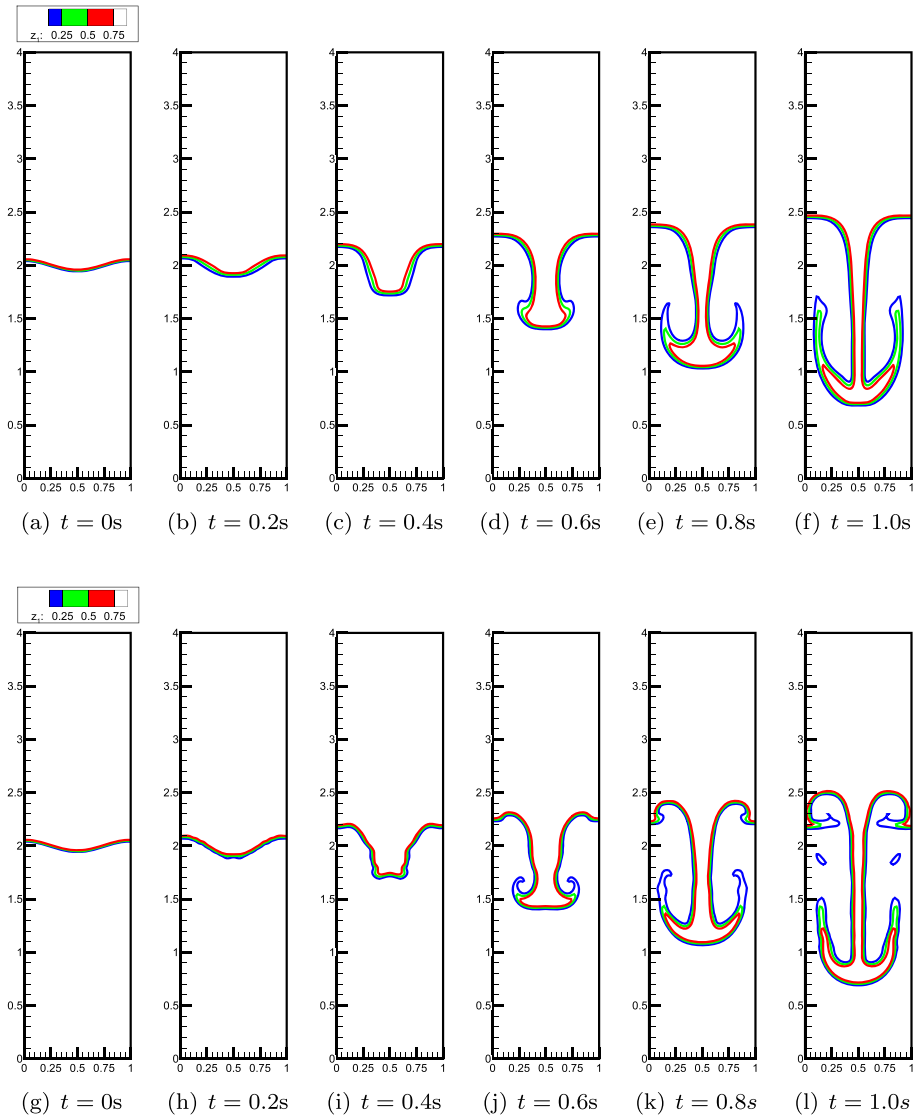
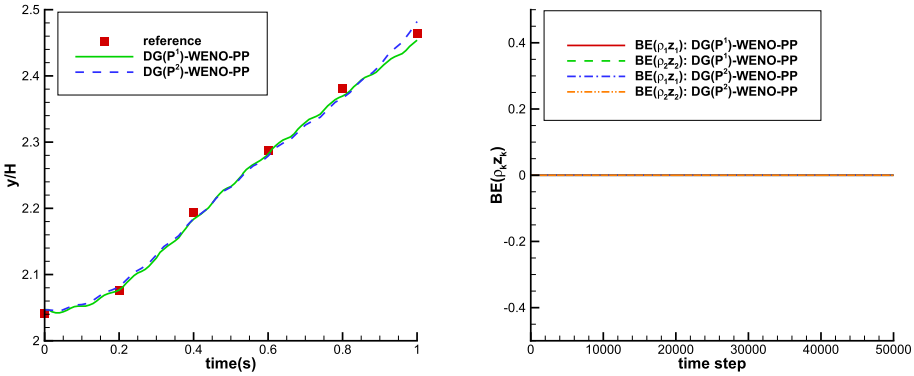


Fig. 10 Interface evolution for the Rayleigh–Taylor instability problem at different time instants (volume fraction contours $z_1 = 0.25, 0.5, 0.75$, top: P^1, bottom: P^2)

The gravity acceleration is $\mathbf{g} = (0, -9.81\ \mathrm{m/s^2})^{\mathrm{T}}$. Free-slip conditions are applied at all boundaries. The speed of sound for both phases are set as $c_1 = c_2 = 100$ m/s and the background pressure is $p_0 = 10{,}000$ Pa.

Figure 10 shows the interface evolution for the Rayleigh–Taylor instability problem plotted by means of the contour lines of $z_1 = 0.25, 0.5, 0.75$ at different time instants solved by the DG-WENO-PP method on a uniform mesh of 64×256 cells. We can see that as time evolves, the heavier fluid starts penetrating into the lighter one under the effect of gravity which results in the initial small perturbation develops into a mushroom-shaped structure. Both of



(a) highest location (b) bound error

Fig. 11 Temporal evolutions of the highest location for the interface and the bound error for the Rayleigh–Taylor instability problem

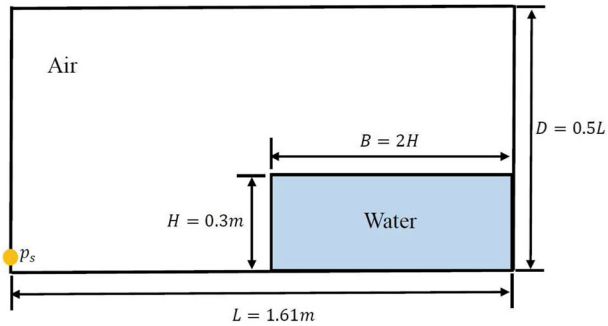
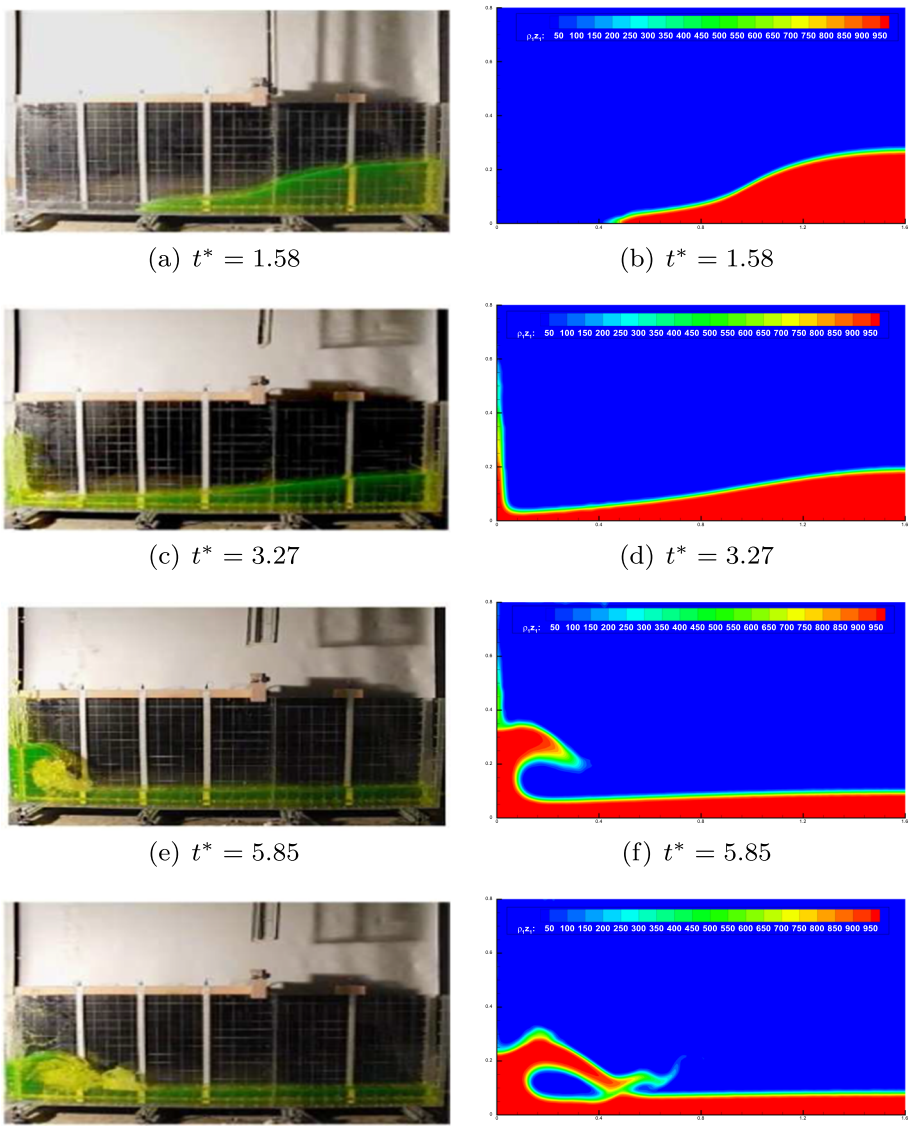


Fig. 12 Schematic illustration of the dam-break flow problem

the DG(P^1)-WENO-PP and DG(P^2)-WENO-PP methods can simulate this process well, while the high-order method can capture more small scale vortex-like structures. Temporal evolution of the highest location for the interface is shown in Fig. 11a which illustrates that the present results are in good agreement with the reference ones [49]. Figure 11b shows the temporal evolution of the global bound error of the partial density which clearly demonstrates the well-maintained positivity-preserving property. Thus, the present DG method is capable of handling two-phase flows involving complex phase interfaces.

5.5 Dam-Break Flow

The last test case is the broken dam-break flow problem which has been widely studied through both experimental [50, 51] and numerical simulation methods [49, 52]. It is used to demonstrate the accuracy and effectiveness of the present method in treating violent free-surface flows. As depicted in Fig. 12, a rectangular water column at rest under the gravity $\mathbf{g} = (0, -9.81\ \mathrm{m/s^2})^{\mathrm{T}}$ with length $B = 0.6$ m and height $H = 0.3$ m is enclosed within an air-filled rectangular container $\Omega = [0, 1.61\ \mathrm{m}] \times [0, 0.805\ \mathrm{m}]$. A pressure sensor P_s is placed at the left wall and its center is 0.015m above the tank bed. Free-slip boundary



(a) $t^* = 1.58$ (b) $t^* = 1.58$

(c) $t^* = 3.27$ (d) $t^* = 3.27$

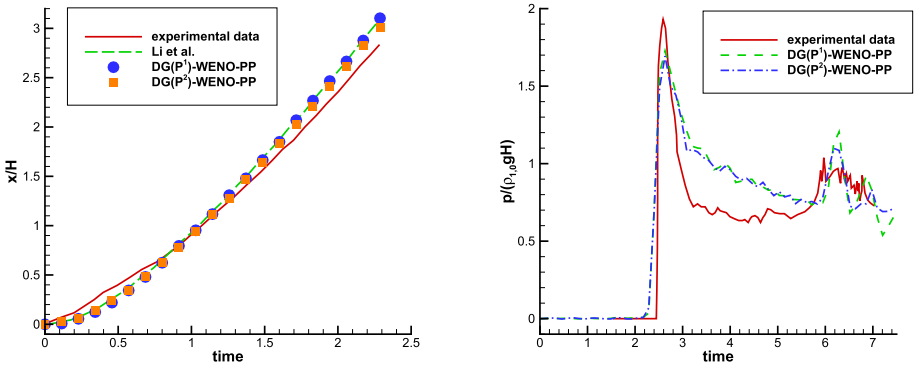
(e) $t^* = 5.85$ (f) $t^* = 5.85$

(g) $t^* = 6.67$ (h) $t^* = 6.67$

Fig. 13 Interface evolution for the dam-break flow problem at different time instants (left: experimental measurements [51], right: DG(P^2)-WENO-PP)

conditions are applied at all the four sides of the rectangular container. The speed of sound for both phases are set as $c_1 = c_2 = 100$ m/s and the background pressure is $p_0 = 10{,}000$ Pa.

In Fig. 13, we display the comparison of flow patterns between the numerical results obtained by the DG(P^2)-WENO-PP method with 128×256 unform mesh cells and the experimental measurements reported in Ref. [51] at four different non-dimensional time instants $t^* = t\sqrt{|g|/H} = 1.58$, 3.27, 5.85 and 6.67. It can be seen that the rectangular column of water first collapses onto the lower surface, then impacts on the opposite solid



(a) wave-front position (b) pressure signal at P_s

Fig. 14 Temporal evolutions of the non-dimensional wave-front position and pressure signal for the dam-break flow problem

wall and finally generates a plunging breaking wave. These detailed phenomenons are well predicted by the DG-WENO-PP method and are coincident with the experimental results.

For a quantitative validation, we measure the non-dimensional surge front and the non-dimensional pressure signal at various time levels and then compare them with the results given in Ref. [24, 51]. As shown in Fig. 14, the computed results are in good agreement with the reference ones. Thus, the present method is able to simulate two-phase flows involving large deformations.

6 Conclusions

The DG approach for simulating weakly compressible two-phase flows is attractive due to its high-order accuracy. However, it is generally hindered by the occurrence of nonphysical values due to the spurious numerical oscillations and the violation of physical constraint. In order to improve the robustness of the high-order DG method and meanwhile dampen the numerical oscillations, a high-order DG method combined with a positivity-preserving limiter and a nonlinear WENO limiter was developed in this work. Moreover, the present method was theoretically proved to satisfy the uniform-pressure–velocity criterion which is a necessary condition for ensuing an oscillation-free phase interface. Several benchmark numerical tests are carried out to validate the accuracy and robustness of the present method in simulating weakly compressible two-phase flows involving large density ratios and complex interface deformations. The present method will be further extended to two-phase flows with more general equations of state in the future work.

Acknowledgements The author acknowledges the support provided by the National Natural Science Foundation of China Nos. 12001020 and 12171046.

Data Availability The datasets generated during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Appendix A

Here, we are going to prove that the WENO reconstructed polynomial solution preserves uniform velocity and pressure. The velocity obtained from the WENO reconstructed solution (36) is

$$u_h^{(r)}(x_i^\alpha, y) = \frac{2u_0\mathcal{W}_r([q_1]_{i,j}) + u_0\mathcal{W}_r([q_2]_{i,l})}{2\mathcal{W}_r([q_1]_{i,j}) + \mathcal{W}_r([q_2]_{i,l})} \equiv u_0,$$
$$v_h^{(r)}(x_i^\alpha, y) = \frac{2v_0\mathcal{W}_r([q_1]_{i,j}) + v_0\mathcal{W}_r([q_2]_{i,l})}{2\mathcal{W}_r([q_1]_{i,j}) + \mathcal{W}_r([q_2]_{i,l})} \equiv v_0.$$

Thus, uniform velocity is preserved. Next, based on the pressure equilibrium

$$p_h^{(r)}(x_i^\alpha, y) = p_0 + c_1^2\left(\frac{(\tilde{\rho}_1)_h^{(r)}}{(z_1)_h^{(r)}} - \rho_{1,0}\right) = p_0 + c_2^2\left(\frac{(\tilde{\rho}_2)_h^{(r)}}{(z_2)_h^{(r)}} - \rho_{2,0}\right),$$

where

$$(\tilde{\rho}_1)_h^{(r)}(x_i^\alpha, y) = \rho_{1,0}\overline{(z_1)}_{i,j} + \frac{\rho_{1,0}}{\rho_{1,0}-\rho_{2,0}}\mathcal{W}_r([q_2]_{i,l}),$$
$$(\tilde{\rho}_2)_h^{(r)}(x_i^\alpha, y) = \rho_{2,0}\overline{(z_2)}_{i,j} - \frac{\rho_{2,0}}{\rho_{1,0}-\rho_{2,0}}\mathcal{W}_r([q_2]_{i,l}),$$

we have the reconstructed equilibrium volume fraction

$$(z_1)_h^{(r)}(x_i^\alpha, y) = \overline{(z_1)}_{i,j} + \frac{1}{\rho_{1,0}-\rho_{2,0}}\mathcal{W}_r([q_2]_{i,l}),$$
$$(z_2)_h^{(r)}(x_i^\alpha, y) = \overline{(z_2)}_{i,j} - \frac{1}{\rho_{1,0}-\rho_{2,0}}\mathcal{W}_r([q_2]_{i,l}),$$

which implies that $p_h^{(r)}(x_i^\alpha, y) \equiv p_0$. This completes the proof.

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